

LOS Assignment - 1

1. What If There Were No Operating System?

Imagine a computer without an operating system. Identify the problems a user would face and investigate how OS features solve them. Demonstrate at least three features using Linux and present a “Without OS vs With OS” comparison

An Operating System is an important system software that acts as a connection between the user and computer hardware. It manages the computer's resources and provides an environment for running applications.

Eg: Linux, Windows and macOS.

Without an operating system using a computer would be very complicated. Users would need to interact directly with the hardware which would be difficult for ordinary users.

Problems Without an Operating System

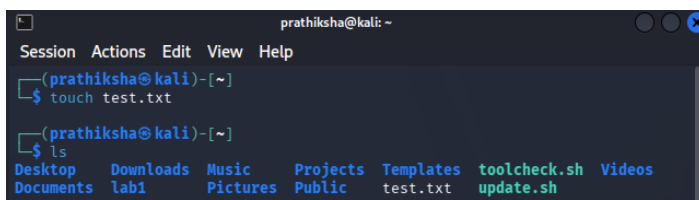
- Running applications would be difficult.
- Users would have to interact directly with computer hardware.
- Creating and managing files would be complicated.
- CPU and memory resources would not be managed easily.
- Devices such as keyboards, printers and USB drives would be difficult to operate.
- Managing users, permissions and security would be difficult.

Without OS	With OS
User must interact directly with hardware	OS provides an interface between user and hardware
Running applications is complicated	Applications can be executed easily
No proper user access control	OS provides users, permissions and security
File management is difficult	Files can be easily created, copied, moved and deleted
Hardware devices must be controlled manually	OS manages hardware devices
Memory and CPU management is difficult	OS manages system resources

Linux Features Demo

Feature 1: File Management

Linux provides commands that allow users to create, view and manage files and directories.



```
prathiksha@kali: ~  
Session Actions Edit View Help  
(prathiksha@kali)-[~]  
$ touch test.txt  
(prathiksha@kali)-[~]  
$ ls  
Desktop Downloads Music Projects Templates toolcheck.sh Videos  
Documents lab1 Pictures Public test.txt update.sh
```

Feature 2: Process Management

Linux provides commands to view the processes currently running on the system.

```
(prathiksha@kali)-[~]
$ ps
  PID TTY          TIME CMD
 6871 pts/0    00:00:00 zsh
 7288 pts/0    00:00:00 ps
```

Feature 3: Hardware and Device Management

Linux provides commands to obtain information about the computer's hardware.

```
(prathiksha@kali)-[~]
$ lsusb
Bus 001 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
Bus 002 Device 001: ID 1d6b:0001 Linux Foundation 1.1 root hub
Bus 002 Device 002: ID 80ee:0021 VirtualBox USB Tablet
```

2. Imagine your team is setting up a cybersecurity laboratory. Explore at least five Linux distributions relevant to cybersecurity and recommend suitable distributions for different roles such as penetration testing, digital forensics, privacy, and security monitoring. Present your findings creatively as a comparison chart, poster, infographic, or “OS selection guide.”

Different Linux distributions are created for different purposes. Some can be used for security testing and research, while others are useful for digital forensics, privacy, system security and cybersecurity learning.

Linux Distribution	Main purpose	Suitable Cybersecurity Role	Important Features
Ubuntu	General purpose Linux	CyberSecurity learning or security lab	Easy to use and widely supported
Kali Purple	Defensive Security	SOC or Security monitoring	Security monitoring and threat detection
Parrot Security OS	Security and Privacy	Security Research	Lightweight security tools
Tails	Privacy	Privacy or Secure browsing	Focuses on anonymity
CAINE	Digital Forensics	Forensic Investigation	Digital evidence analysis

1. Ubuntu

Ubuntu is a popular and user-friendly Linux distribution. It can be used to set up a cybersecurity laboratory and learn Linux commands, system administration and basic security concepts.

It is also useful because many cybersecurity tools can be installed and tested on Ubuntu.

Best suited for:

- Cybersecurity Learning
- Security Lab Setup
- Linux Administration

2. Kali Purple

Kali Purple is a security-focused version of Kali Linux that emphasizes defensive security. It provides an environment for security monitoring, detection and analysing suspicious activity.

It is useful for understanding how security teams monitor systems and respond to possible threats.

Best suited for:

- Security Monitoring
- SOC
- Threat Detection
- Defensive Security

3. Parrot Security OS

Parrot Security OS is a Linux distribution designed for cybersecurity, privacy and security research. It provides various tools that can be used for security testing and analysis.

It is also relatively lightweight, making it suitable for students and security researchers.

Best suited for:

- Security Testing
- Security Research
- Privacy

4. Tails

Tails is a Linux distribution mainly designed for privacy and anonymity. It can be started from a USB drive without permanently installing it on the computer.

It is useful when protecting personal privacy and maintaining anonymity while using the internet.

Best suited for:

- Privacy
- Anonymous Browsing
- Secure Temporary Sessions

5. CAINE

CAINE stands for Computer Aided INvestigative Environment. It is a Linux distribution designed mainly for digital forensics and cyber investigations.

It provides tools that help investigators collect, examine and analyse digital evidence from computers and storage devices.

Best suited for:

- Digital Forensics
- Evidence Analysis
- Cyber Investigation

3. Exit status:

Execute at least five Linux commands that produce both successful and unsuccessful results.

Investigate their exit status using ` \$?`.

Create a simple visual/table showing Command → Result Exit Status → Meaning.

What can a script learn from an exit status?

In Linux each command gives an exit status after it is executed. This status shows whether the command was completed successfully or if an error occurred.

0 - Command was successful

Non-zero value - Command failed or an error occurred

The special variable \$? is used to check the exit status of the last executed command.

1. Successful Command

```
(prathiksha@kali)-[~]  
$ pwd  
/home/prathiksha  
  
(prathiksha@kali)-[~]  
$ echo $?  
0
```

2. Successful command

```
(prathiksha@kali)-[~]  
$ ls  
Desktop  Downloads  Music  Projects  Templates  toolcheck.sh  Videos  
Documents  lab1  Pictures  Public  test.txt  update.sh  
  
(prathiksha@kali)-[~]  
$ echo $?  
0
```

3. Unsuccessful Command

```
(prathiksha@kali)-[~]  
$ ls xx88  
ls: cannot access 'xx88': No such file or directory  
  
(prathiksha@kali)-[~]  
$ echo $?  
2
```

4. Successful Command

```
(prathiksha@kali)-[~]  
$ mkdir testfolder  
  
(prathiksha@kali)-[~]  
$ echo $?  
0
```

5. Unsuccessful Command

```
(prathiksha@kali)-[~]  
$ cat lmn.txt  
cat: lmn.txt: No such file or directory  
  
(prathiksha@kali)-[~]  
$ echo $?  
1
```

Command and Exit Status

Command	Result / Exit Status	Meaning
pwd	0	Command executed successfully
ls	0	Files and directories displayed successfully
ls xx88.txt	2	The specified file or directory was not found
mkdir testfolder	0	Directory created successfully
cat lmn.txt	1	The specified file could not be found

A script can use the exit status to check whether a command was successful or failed. It helps the script detect errors and decide whether to continue with the next step or stop the execution.